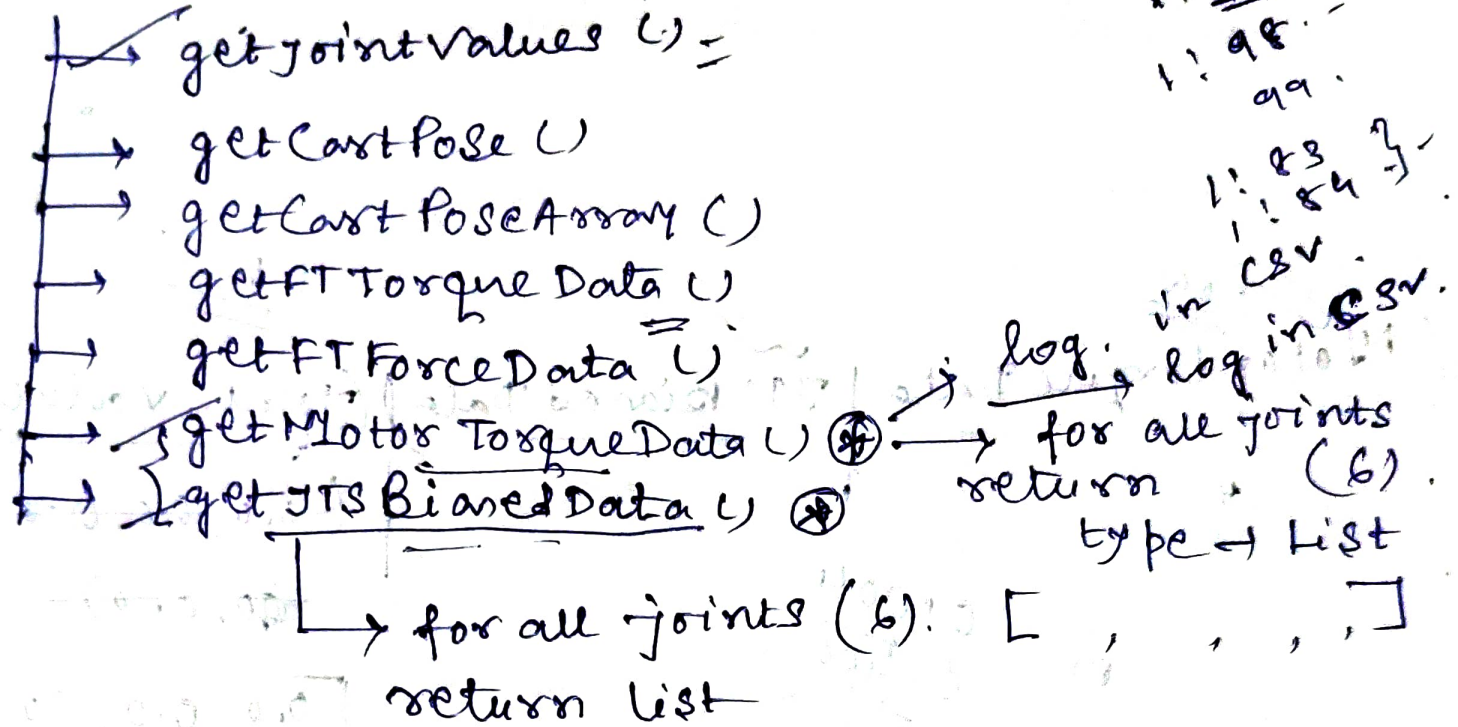
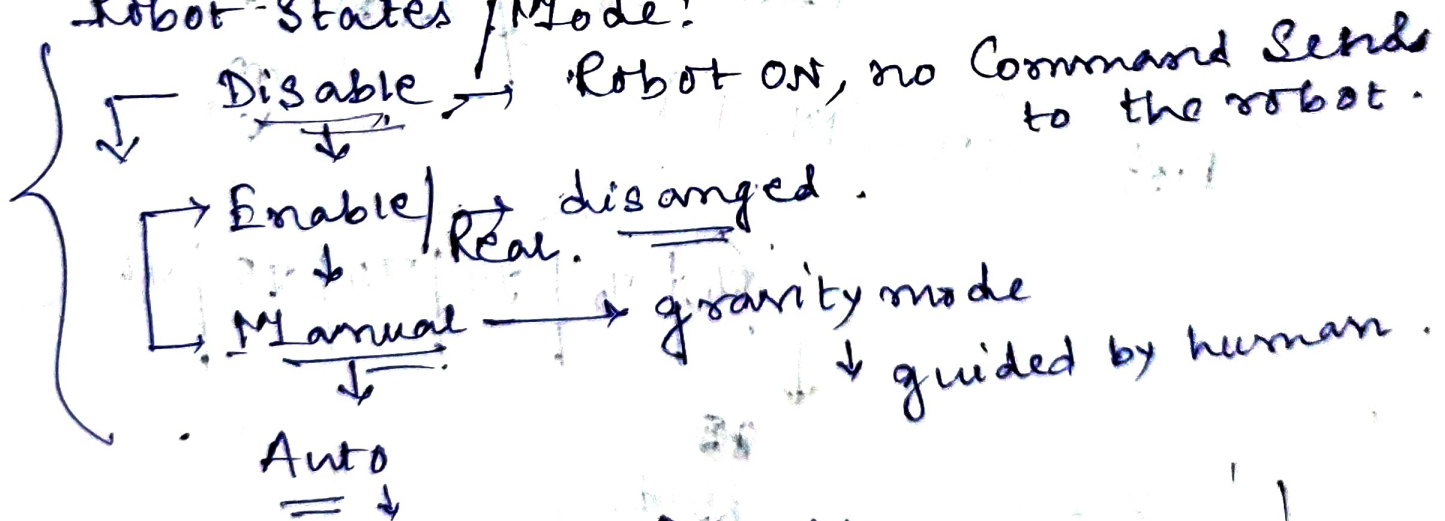


21/05/26

Feedback and Sensing Command



Robot States / Mode:



Conditions.

- i) Disable →
- ii) Real / Manual

3 (Actually → Real / Manual)

③

②

(5-6)

<u>Row</u>	<u>Motor torque Data</u>	<u>JTS Biased Data</u>	<u>Joint-values</u>
4			(0.0, 0.0, 0.0) (90, 0.0)

$$D_1 = \frac{0.004}{0.400}$$

[0.0, 0.0, 90.0,
0.0, 0.0, 90.0]

gear-ratio = $\frac{\text{JTS Biased Data}}{\text{Motor torque Data}}$

≠ Const. ↓
m

$$\text{Diff} = (\text{JTS} - \text{ML-T})$$

Offset

1: 85
1: 120
1: 80

↓

Ⓜ	J ₂	J ₃	J ₄	J ₅	J ₆
85					
85.71					
120.0					
80.14					

20/05/26 → Friction Compensation Model [Find parameters] ⁽³⁾

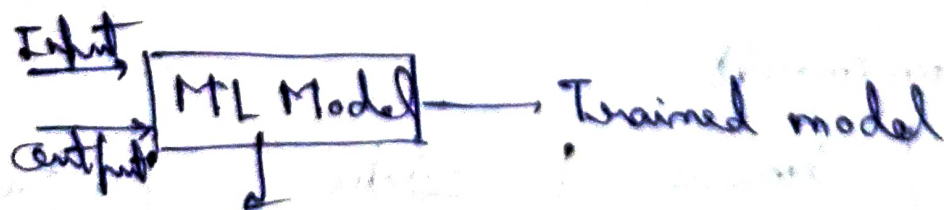
↓ → 6 DOF, Robotic arm
1 May to 20 July ✓
↳ Study → Theory of m/cs & mecha
↳ D-H param
↳ Robotics Basics
↳ ~~Forward~~ Jacobian's eq. criterion.

Work:- 6 DOF → 6 Joints

↓
Find ~~friction~~ friction

To do ML model to find friction model to define the parameters.

20/05/26



Read Physics Based ML (Reference model of fric to give fric model)
→ Find mathematical models. relations.

Parameters calculated from 6 DOF,

Q

T $\rightarrow$ T_{Motor} , T_{Joint}

w

26/05/26

Friction models explained

$\rightarrow$ 2 rough surfaces touch each other.

$\rightarrow$ Surfaces are not smooth. [Tiny lumps - Asperities]

force applied $\rightarrow$ tiny lumps ^{binds} ~~breaks~~ $\rightarrow$ called stick
" high $\rightarrow$ " " breaks $\rightarrow$ " slip.

Simple friction models:-

<u>Coulumb</u>	<u>Viscous</u>	<u>Static</u>	<u>Stiction effect</u>
const fric off to motion	$F_f \propto vel$	more heavy friction, at first it did not move.	vel $\uparrow$, fric $\downarrow$ then inc. due to lubricants.
start motion $\times$ stop " $\times$ zero vel $\times$			

(5)

Pre-sliding Displacement :- Before sliding starts, surfaces actually deform like springs.

Frict memory :- Frict does not react instantly, it remembers previous ~~point~~ motion.

Complex fric model :-

7-param model :-

- ↳ Coulumb, Viscous, Static fric, ~~break~~
- ↳ Stiction, memory, pre-sliding ~~off~~
- ↳ Complex and need many parameters.

Static variable fric model :- No need.

Karnopp model :- Solve prob near zero vel.

- ↳ Sys treated as stopped.
- ↳ easy for computer simul.

Mod Dahl model :- Stiction off

- ↳ Hysteresis behave / Vel reversal off

M2 model :- Adv dynamic fric model

- ↳ Sticking / sliding fric.

Lugre Model :- Surface contact $\rightarrow$ tiny elastic models.

$\hookrightarrow$ avg hysteresis deformation.

- Can model
- $\rightarrow$ Stick-slip
 - $\rightarrow$ free sliding
 - $\rightarrow$ stick break off
 - $\rightarrow$ dynamic friction.

Simple :- ~~is a simple~~ See only for Lugre Model

- $\rightarrow$ Stick slip motion
- $\rightarrow$ thermal
- $\rightarrow$ Spring like sticking.

Advantages and Disadvantages :-

- $\rightarrow$ High accuracy
- $\rightarrow$ Simple model
- $\rightarrow$ Contact application
- $\rightarrow$ Complexity allowed.

Main idea: Fric opposes motion, causes vibration, inaccuracy, energy loss, jerky motion (stick/slip)

- 1) Coulumbi fric model ✓
- 2) Viscous " ✓
- 3) Stiction off ✓
- 4) Dahl model: - ~~Two~~ Tiny surfaces behave like springs
 - ↳ fore-sliding motion
 - ↳ Hysteresis.
- 5) Lugre Model ✓
- 6) Lieners " :- Improves Lugre Model by handling
 - ↳ Hysteresis memory
 - ↳ Vel reversals more acc
 - ↳ Maths harder.
- 7) Generalised Maxwell model :- Many spring elem together
 - ↳ Very acc
 - ↳ Matches real fric behave.
 - ↳ High computational complexity.

Friction compensation :- have to reduce fric effects.

Feed-Forward

- feed fric before,
- add offset control force
- fast response,
- high acc.
- needs acc fric model.

Feed-Back

- Detect error continuously
- correct motion in real time
- robust against disturbances
- combine with feed fwd.

RESEARCH APPROACH

from papers 2005, 2024, 2024, 2026.

- ⇒ Fric identification (2005) → LuGre + Maxwell slip and hysteresis modelling.
- ⇒ Learning transferable (2024) → Phys inform neural networks for LuGre param & transferable fric.
- ⇒ H based adaptation (2024) → Residual learning fric adaptation for robotic joints.
- ⇒ Res on param iden (2024) → PINN based param for LuGre / Strickland

(9)

Possible joint estimation idea

$$\tau_{\text{pic}} = \tau_{\text{motor}} - \tau_{\text{joint}}$$

→ conditionally correct.

Complete vel is, $\tau_{\text{motor}} = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + \tau_f + \tau_{\text{ext}}$

joint input desired, $\tau_f + \tau_{\text{ext}}$

Mass acc + Coriolis + Gravity
pic

$$\therefore \text{pic torque, } \tau_f = \tau_{\text{motor}} - [M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)]$$

↓
Under no ext disturbances ($\tau_{\text{ext}} = 0$).

But, during acc → inertia torque contaminates pic estimation

" high speed motion → Coriolis eff occur

" changing poses → gravity torque changes

" dynamic trajectories → subtracting motor $\times$ joint torque overestimates pic.

My idea works for → Const vel tests,

low acc,

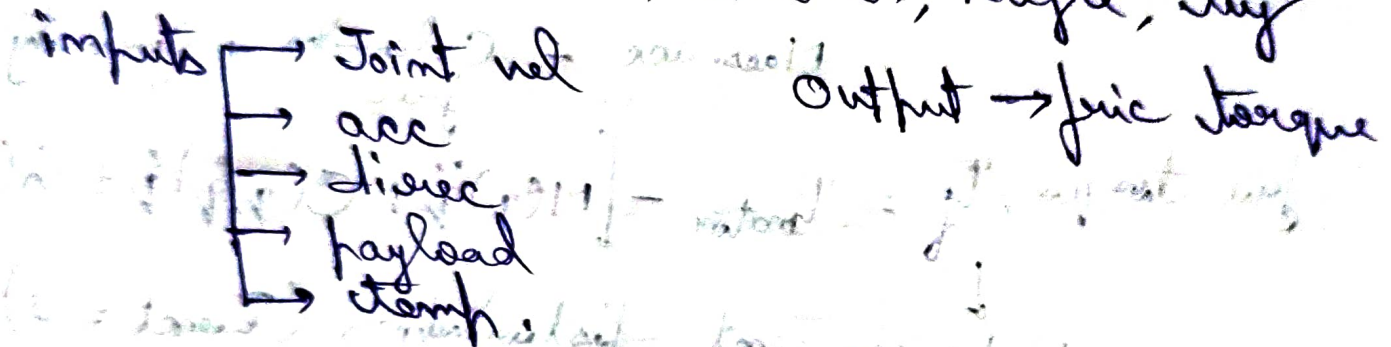
low coriolis coupling,

quasi-static motion.

Architecture :-

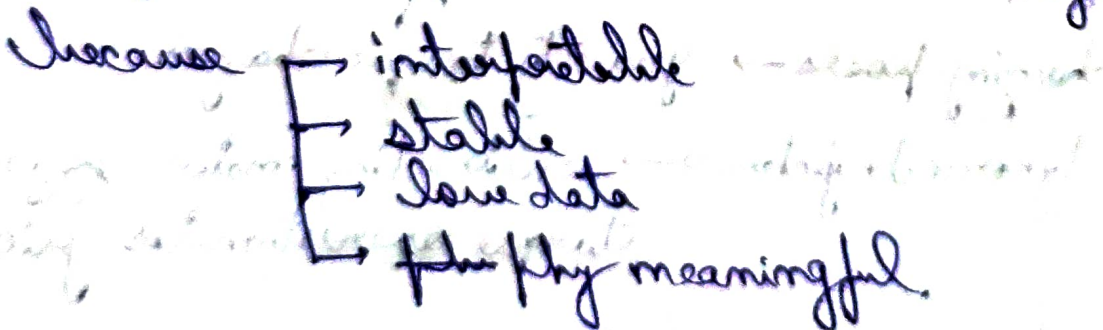
S1) Classical fric identify

→ Coulumb, Viscous, Steinhilber, LuGre, Jay



S2) Physics based Baseline model.

LuGre / Steinhilber as baseline → $T_{physics}$



S3) ML Residual learning.

Train ML on residual, $T_{res} = T_{real} - T_{phy}$

Final model → $T_{total} = T_{phy} + T_{ML}$

11

- less data req.
- phy constrained
- better extrapolation
- more stable in real time
- easier deployment.

ML Inputs:-

Priority	Feature	Why?
1	Angular ($\dot{q}$)	Core fric dependant.
2	Vel sign	Dirac asymm
3	Reverse history	Hysteresis / Stick Slip
4	Acc ($\ddot{q}$)	Dynamic fric eff.
5	Load/gravity torque	Load-def fric.
6	Temp	Lubricant viscosity viscosity changes
7	Reverse fric	Dynamic memory
8	Position (q)	Harmonic drive Periodic effs.
9	Motor current	Torque feedback
10	Time window features	Dynamic behaviour

velocity input feature, all classical models are def.

$$\Rightarrow \tau_f = f(\dot{q})$$

Coulumb $\rightarrow \tau_f = F_c \cdot \text{sgn}(\dot{q})$

Viscous $\rightarrow \tau_f = B \cdot \dot{q}$

Stirchek $\rightarrow \tau_f = \left(F_c + (F_s - F_c) e^{-|\frac{\dot{q}}{v_s}|^\delta} \right) \text{sgn}(\dot{q}) + B \dot{q}$

Lugre $\rightarrow \dot{z} = \dot{q} - \frac{\sigma_0 |\dot{q}|}{F_c + (F_s - F_c) e^{-|\frac{\dot{q}}{v_s}|^\delta}} \cdot z$

Revolute elastic joint, angular velocity is suff.

$$v = r \cdot \omega \rightarrow r = \text{constant.}$$

$\therefore$ pic models can use ang vel (ω).

Best ML models:-

Random forest $\rightarrow$ Non linear static pic.

XG Boost $\rightarrow$ Excellent baseline.

ANN $\rightarrow$ Complex non linear mapping.

LSTM $\rightarrow$ Best for hysteresis / dyn pic

PI NN $\rightarrow$ Best over level off.

Long Short
Term Memory
Recur NN.

Best Aff

- Staircase baseline
- Lugre dyn model
- XGBoost residual model
- LSTM for dynamic "
- PINN if time permits.

(13)

Experiments :-

Const Vel test

- find →
- Coloumb fric
 - Vis "
 - staircase off.

- steps →
- run each joint separately
 - fixed vel
 - both directions
 - multiple payloads

Plot → T_f vs $\dot{q}$

- staircase tip near zero.
- asym +ve/-ve fric.

Low speed stick-slip test

imp for lugre validation

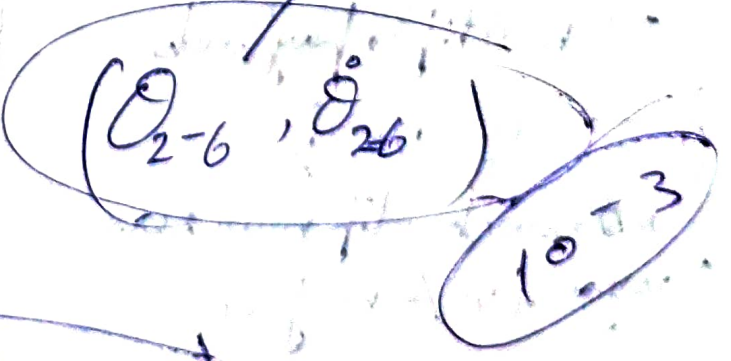
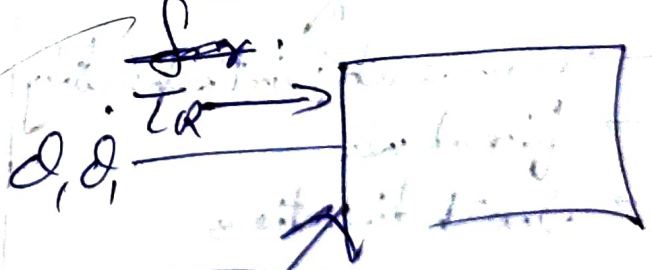
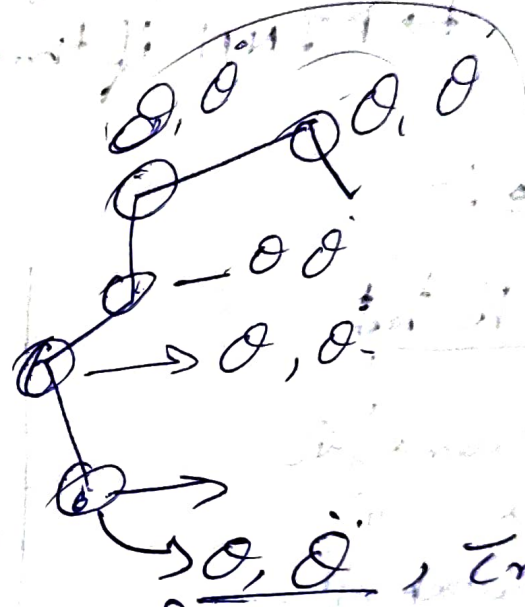
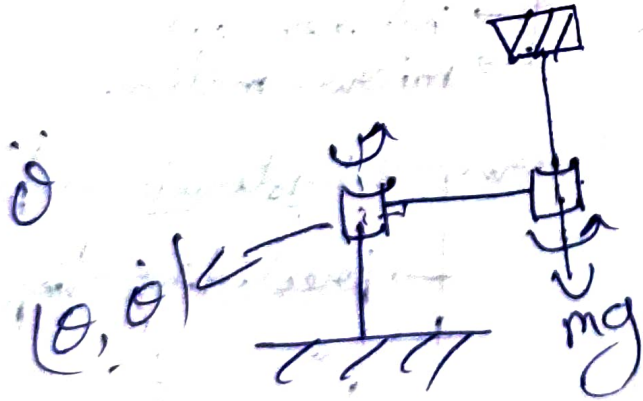
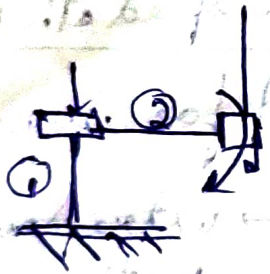
- Use →
- ultra low sinusoidal motion
 - vel reversal
 - micro-motion.

- observe →
- hysteresis loops.
 - presliding slip
 - fric lag.

Compensation Torque :-

$$T_{\text{comp}} = T_{\text{nominal}} + \tau_{\text{fric-estimated}}$$

14



$$\tau_s = f_1(q_1, \dot{q}_1) + f_2(q_{2-6}, \dot{q}_{2-6})$$

$$a\ddot{\theta}_2 + b\dot{\theta}_2 + c\theta_3$$

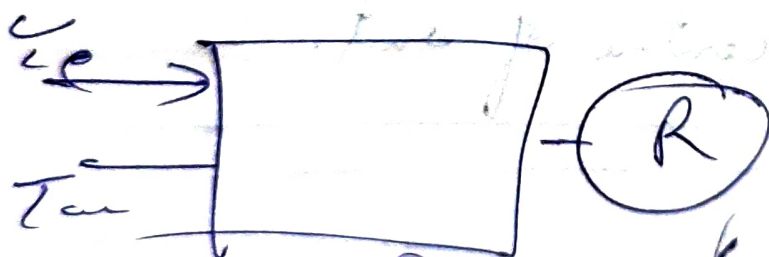
(15)

$$m\ddot{a} + c\dot{a} + g(a) + \underline{f_s} = \underline{\tau}$$

$$\underline{\tau_{mo}} = \underline{f(\vartheta, \dot{\vartheta})} \quad [f_c, f_s \in \text{Cont}]$$

$$\tau_{exp} = \tau_c$$

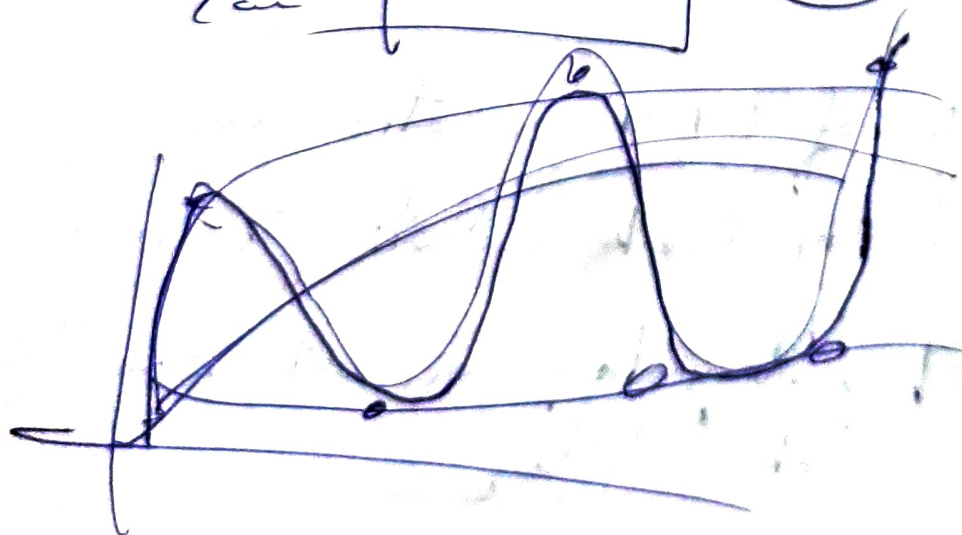
$$\tau_c = \tau_m = \tau_R$$



$$f(\vartheta) = \underline{a + b\vartheta}$$

$$\underline{a, b}$$

$$a + b\vartheta + c\dot{\vartheta}$$



Frictional torque = Fric force equivalent to translational motion.

$$\vec{T}_f = \vec{F}_f \times \vec{r}$$

- Spinning disc / rot shaft $\Rightarrow$ fric does not act at one point. It acts as a tangential force distributed along contact surface.
- To calculate total fric force/torque acting on a system, integrate fric force over entire contact area, multiply by dist from centre of rot.

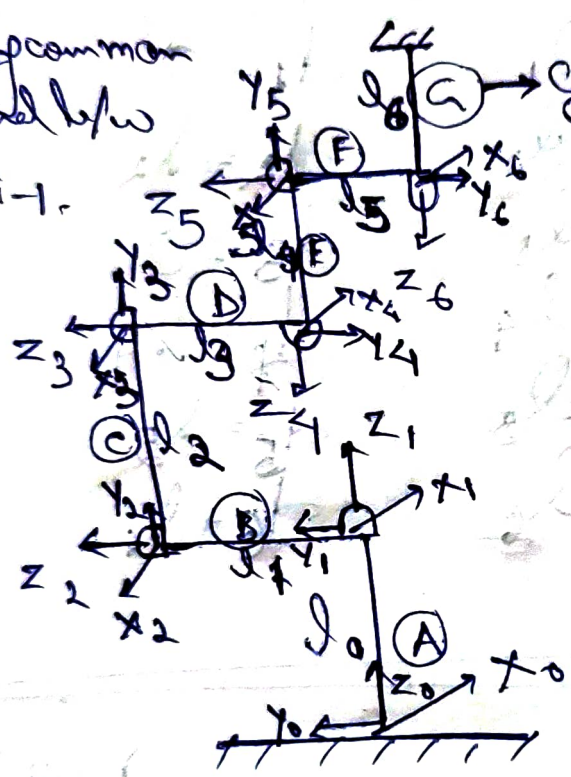
01/06/26

Tasks

- ① D-H Parameter of 6 DOF robotic arm
- ② Matlab Peter Corke robotic toolbox
- ③ Idea of seq, trajectory robot should follow to take data.

gear ratio = 100:1 [Each joint gear drive]

$X_{i-1} \rightarrow$ Along common normal b/w $Z_i \cancel{Z_{i-1}}$



Home Configuration

Link no $\rightarrow$ 0 to n

Joint no $\rightarrow$ 1 to n

' n ' DOF = $(n+1)$ links including fix link (base/link 0).

Link length (a_i) $\rightarrow$ Dist b/w $Z_i \cancel{Z_{i-1}}$ along X_i axes

" twist (α_i) $\rightarrow$ Angle b/w $Z_i \cancel{Z_{i-1}}$ about X_i in R.H.

Joint length (d_i) $\rightarrow$ Dist " $X_{i-1} \cancel{X_i}$ along Z_i -axis

" twist (θ_i) $\rightarrow$ Angle " " " about Z_{i-1} R.H.

$$\alpha_1 = 0^\circ$$

$$\alpha_2 = +90^\circ$$

$$\alpha_3 = 0$$

$$\alpha_4 = -90^\circ$$

$$\alpha_5 = +90^\circ$$

$$\alpha_6 = -90^\circ$$

$$\alpha_7 =$$



l_0

l_1

l_2

l_3

l_4

l_5

l_6

$$d_1 = l_0$$

$$d_2 = 0$$

$$d_3 = l_2$$

$$d_4 = 0$$

$$d_5 = l_4$$

$$d_6 = l_5$$

$$d_7 = l_6$$

⑧

Q_1

Q_2

Q_3

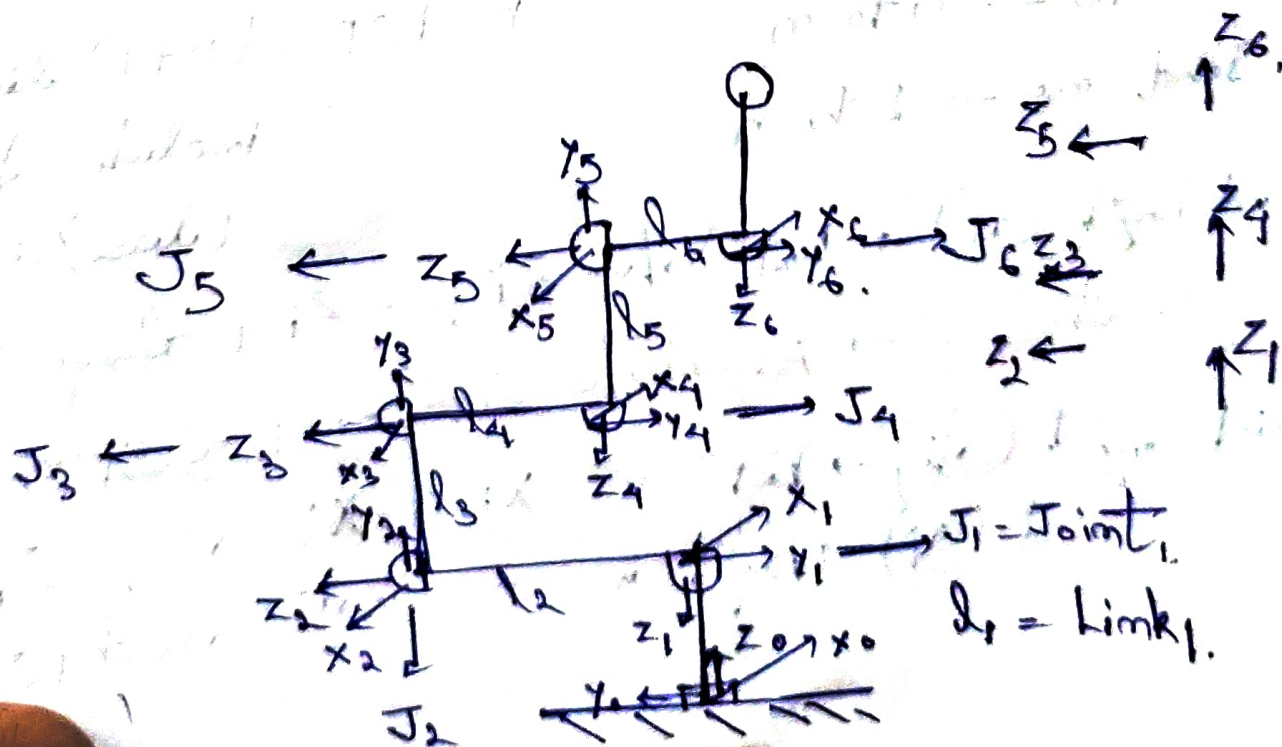
Q_4

Q_5

Q_6

Q_7

D-H Parameter



$$\alpha_1 = +180^\circ$$

$$\alpha_2 = +90^\circ$$

$$\alpha_3 = 0^\circ$$

$$\alpha_4 = -90^\circ$$

$$\alpha_5 = +90^\circ$$

$$\alpha_6 = -90^\circ$$

$$\alpha_7 =$$



$$a_1 = 0$$

$$a_2 = 0$$

$$a_3 = d_3$$

$$a_4 = 0$$

$$a_5 = 0$$

$$a_6 = 0$$



$$l_1 = +l_1$$

$$l_2 = +l_2$$

$$l_3 = 0$$

$$l_4 =$$

$$l_5 =$$

$$l_6 =$$

$$l_7 =$$

$$l_8 =$$

$$l_9 =$$

$$l_{10} =$$

(19)



Math Lab

○ → ✓

α → ✓

α → Joint list (l)

d → Link length (a_i)

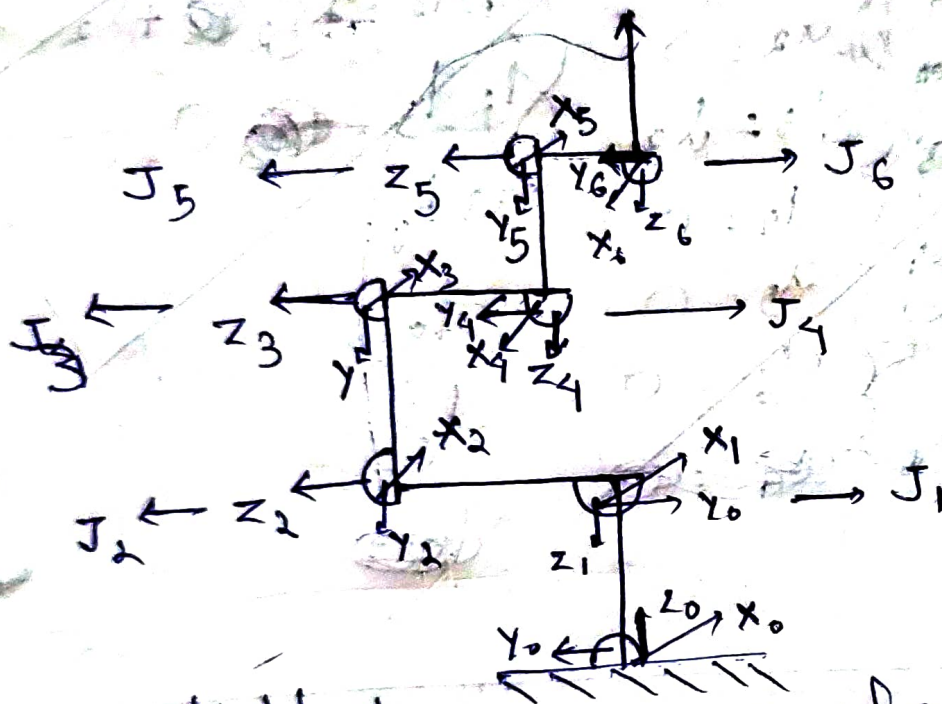
6 DOF → All revolute joint

> No prismatic ...

↙ → Revolute

|| → Prismatic.

(20)



All link lengths = l .

ACW = +
CW = -

Link length

Link twist angle

Joint distance

Joint angle

$$a_1 = 0$$

$$\alpha_1 = 0$$

$$d_1 = l$$

$$\theta_1$$

$$a_2 = 0$$

$$\alpha_2 = -90^\circ$$

$$d_2 = l$$

$$\theta_2$$

$$a_3 = +l$$

$$\alpha_3 = 0$$

$$d_3 = l$$

$$\theta_3$$

$$a_4 = 0$$

$$\alpha_4 = +90^\circ$$

$$d_4 = l$$

$$\theta_4$$

$$a_5 = 0$$

$$\alpha_5 = -90^\circ$$

$$d_5 = l$$

$$\theta_5$$

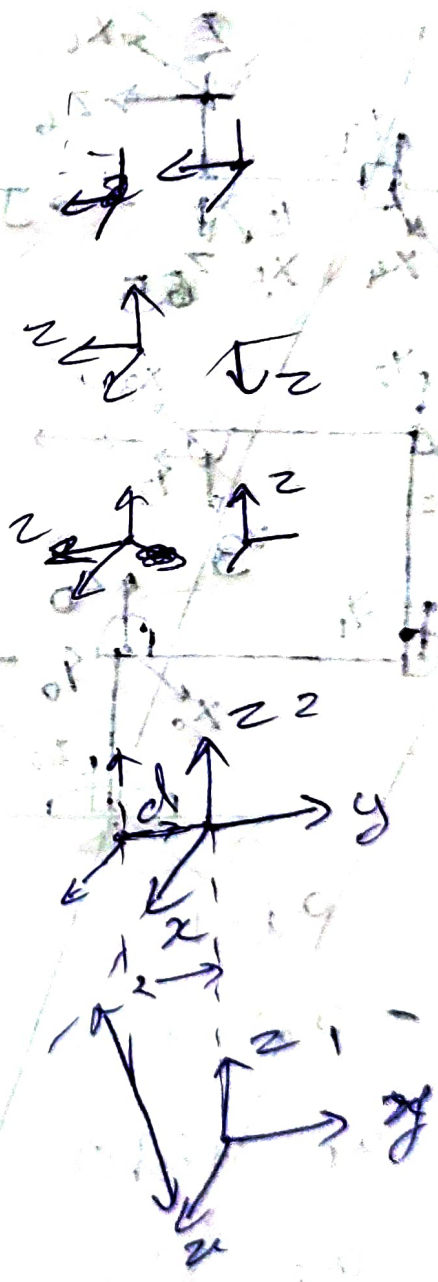
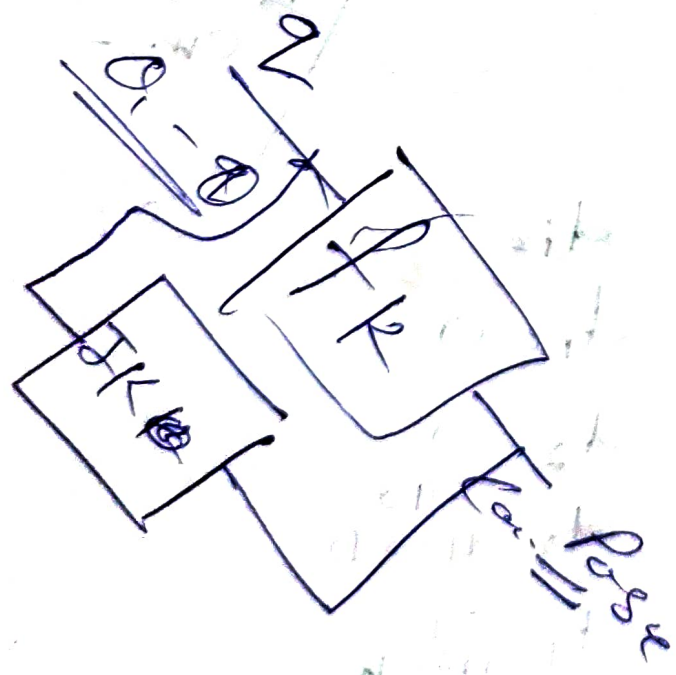
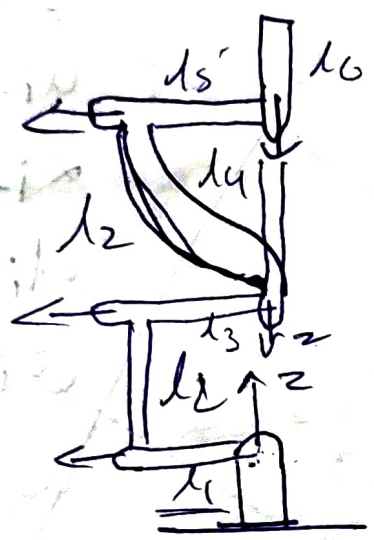
$$a_6 = 0$$

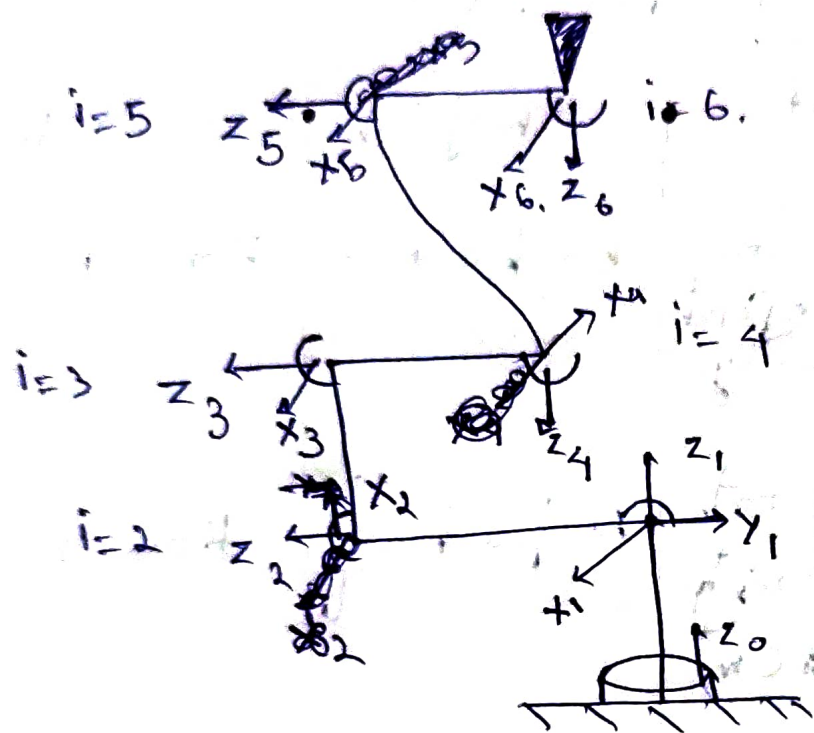
$$\alpha_6 = +90^\circ$$

$$d_6 = l$$

$$\theta_6$$

(21)





$$a_{i-1}$$

$$\alpha_{i-1}$$

$$d_i$$

$$Q_i$$

$$ACW = +$$

$$CW = -$$

$$\frac{\text{Link twist}}{\alpha_{i-1}}$$

$$\frac{\text{Link length}}{a_{i-1}}$$

$$\frac{\text{Joint length offset}}{d_i}$$

$$\frac{\text{Joint angle}}{Q_i}$$

$$\alpha_{0(i=1)} = 0$$

$$a_0 = 0$$

$$d_1 = 163.13$$

$$Q_1$$

$$\alpha_1(i=2) = +90^\circ$$

$$a_1 = 0$$

$$d_2 =$$

$$Q_2$$

$$\alpha_2(i=3) = 0$$

$$a_2 = +390$$

$$d_3 = 0$$

$$Q_3$$

$$d_4 = \frac{163.13}{+390}$$

$$Q_4$$

$$d_5 =$$

$$Q_5$$

$$d_6 = \frac{163.13}{1390}$$

$$Q_6$$

$$+960$$

$$\alpha_5(i=6) = +90^\circ$$

$$a_5 = 0$$

Exp List (Payload = 0)

(24)

Exp No	Joint no	Joint motion	other joints	Orientation
1-6	J ₁ /2/3/4/5/6 move individually	Keep others fix Move each joint individually	Keep others fix	Home Pos.
7	J ₁ /2/3/4/5/6 vel off study	Move each joint at diff velocity (CW and ACW)	Keep others fix	Home Pos.
8	Direction Study			
9	Gravity off	Move only gravity	others fix	Home Pos.
10	Coupling off	J ₁ +2/J ₂ +3/J ₃ +4/ J ₄ +5/J ₅ +6 J ₁ +3/J ₁ +4/J ₁ +5/J ₁ +6 J ₂ +4/J ₂ +5/J ₂ +6 J ₃ +5/J ₃ +6 J ₄ +6.	Others fix	Home Pos/ Any other Pos.
11	Full trajectory	Diff joints move to diff trajectory in work space	Maybe one/two fix or may be next	Pick & Place

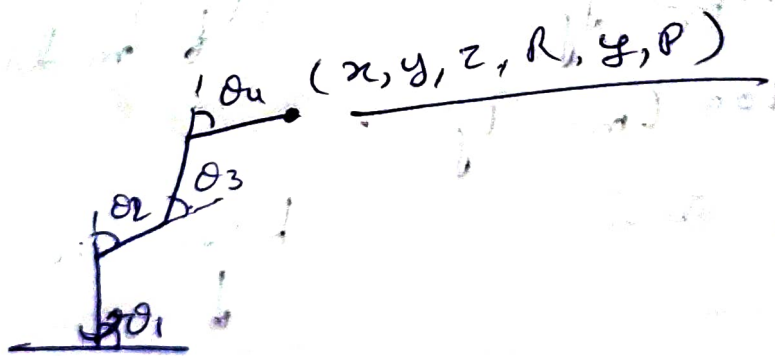
23

$$Q = (5\%, 10\%, 20\%, 50\%, 100\%) \text{ } Q_2$$

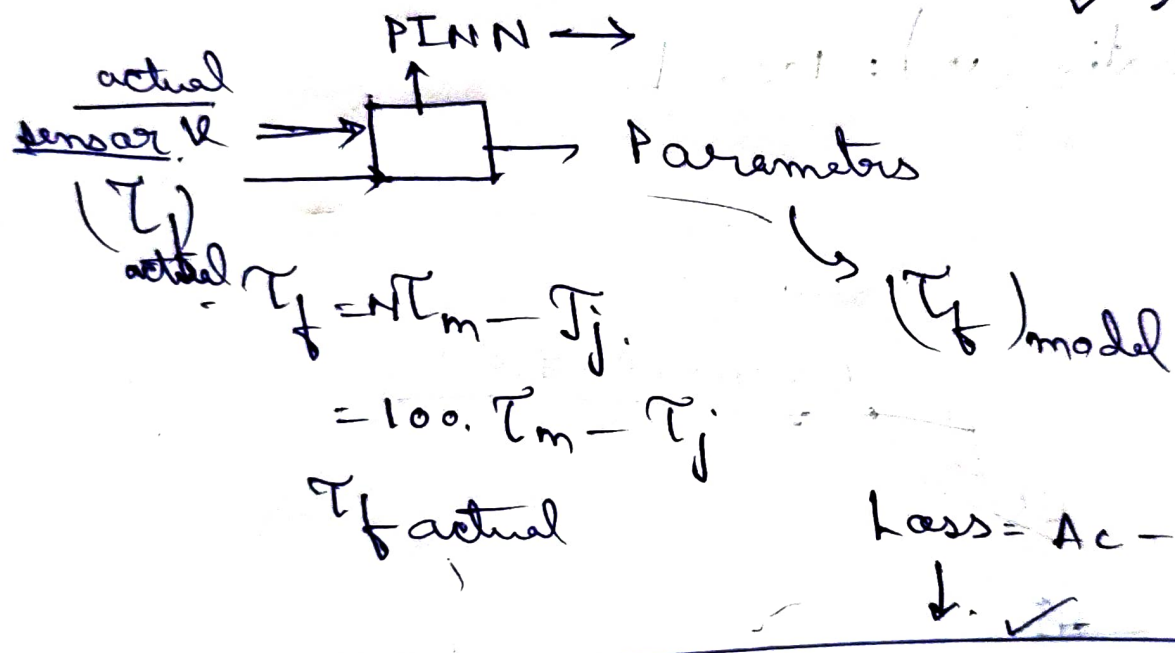
repetitions = 5 times

CW / ACW,

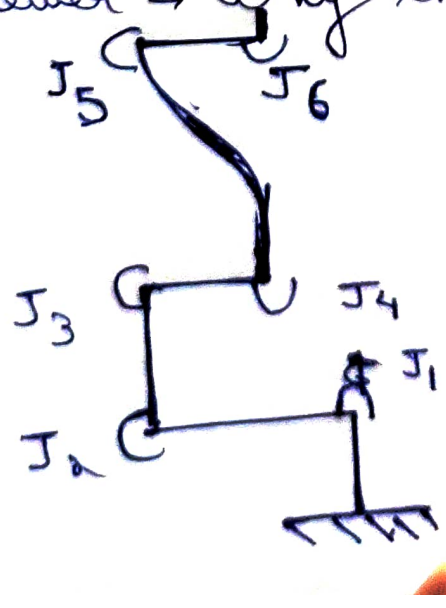
Gear ratio (r) = 100 : 1



$$f_{12}(a_1 - a_6) = (x, y, z, R, yP)$$



- Trajectory selection planning acc to papers see
- Reviewer → why this trajectory not others.



- 1) Move condition, with known positions (During runtime)
 $\rightarrow [-71.82, -15.76, 88.39, -12.12, 64.05, -38.91]$

At real cond $\rightarrow$

Command $\rightarrow$ rel in $\times$ acc (initial) $\rightarrow 30 \text{ rad/s}$
 30 rad/s^2

- 2) Static conditions:

$\rightarrow$ [Same Pos]

JTS $\overline{\text{|||||}} \quad \overline{\text{|||||}}$
 |||||

Ptest - 1

move move $\rightarrow$

$\begin{matrix} J_s - 30 \\ - 6 \\ - 50 \end{matrix} \left. \vphantom{\begin{matrix} J_s - 30 \\ - 6 \\ - 50 \end{matrix}} \right\} \rightarrow \text{alat} - \begin{matrix} -90 \\ +90 \end{matrix}$



$\rightarrow$ go to move

move in 4

$\begin{matrix} \text{8 cm} \\ \text{I} \end{matrix} \left. \vphantom{\begin{matrix} \text{8 cm} \\ \text{I} \end{matrix}} \right\} -170 + 180$

given

03/06/26

EXP STARTS

(28)

Exp - 1 → Home Pos [File name
Date - Data - Exp 1]

81) Run robot from disable to enable

82) In Home Pos only

Collect data

task 5 → Joint 5 → vel Angle
30
60
90



data

Time Stamp	J ₁	J ₂	...	J ₅	JTS	1	...	5	vel	1	3	5	Pos	1	...	3	5
→																	
→																	

task 4 → Joint

task-no = 1

1 disable
2 ↓
1 Enable
↓
Real

(Stary.)

J-1

if task-no = 1
J₁ = (

Time-1 MTS(6) JJS(6) O(6) O(6) 2(6)
 J_1, J_2, J_3

move joint $\rightarrow$ 5 to 1
 joint-move () forward
 joint-vel ()
 goto Home ()

Combination - 1
 for
 vel $\rightarrow 30 \text{ rad/s.}$
 acc $\rightarrow 30 \text{ rad/s.}$

reverse
 J-
 J-vel ()
 - goto home ()

Combination - 2
 for vel $\rightarrow 45 \text{ rad/s.}$
 acc $\rightarrow 45 \text{ rad/s}^2$

Combination - 3
 for - vel $\rightarrow 60 \text{ rad/s.}$
 acc $\rightarrow 60$

N.B $\rightarrow$ exception in joint-2
 Set vel only 30 rad/s.
 " acc " 30 rad/s.

Max Range

Start

Max

Min

J	<u>Max</u>		<u>Min</u>	
	α (degree)	ψ (degree/1/s)	α (degree)	ψ (degree/1/s)
J ₁	+180	60	-180	
J ₂	95/70	60	-95/70	
J ₃	160/100	60	-160/100	
J ₄	180	60	-180	
J ₅	160	60	-160	
J ₆	180	60	-180	

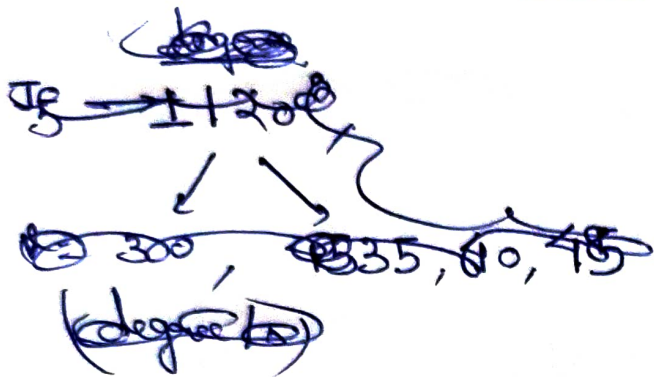
$\psi \rightarrow 30, 45$

hing $\rightarrow 192.168.20.203$

for (

Data

(31)



In loop,
5 times.

$$J_4 \rightarrow \pm 120^\circ$$

~~300~~ ~~335~~ ~~40~~ ~~45~~

$$V = \text{30, 32, 33, 36, 40, 45}$$

30, 32... (46)

$$J_5 \rightarrow \pm 100^\circ$$

$$V \rightarrow \text{30, 33, 36, 40, 45}$$

30, 32... 46

$$J_6 \rightarrow \pm 120^\circ$$

$$V \rightarrow \text{30, 33, 36, 40, 45}$$

30, 32... 46

$$\text{Max } V = 46$$

$$J_2 \rightarrow \pm 60^\circ$$

$$V \rightarrow 30 \text{ degree}/\Delta$$

$$J_3 \rightarrow \pm 100^\circ$$

$$V \rightarrow 30, 40$$

$$J_1 \rightarrow \pm 150^\circ$$

$$V \rightarrow \text{30, 33, 36, 40, 45}$$

30, 32... 46

Harmonic drive.

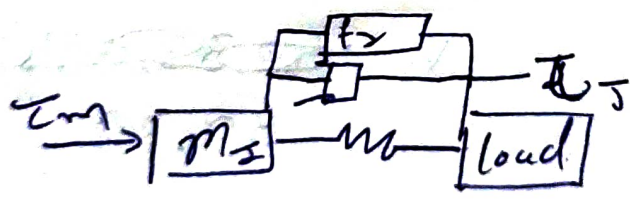
~~Spring constant.~~

→ Inertia

→ Spring

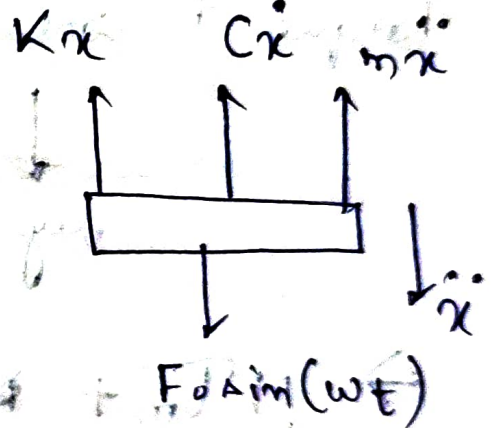
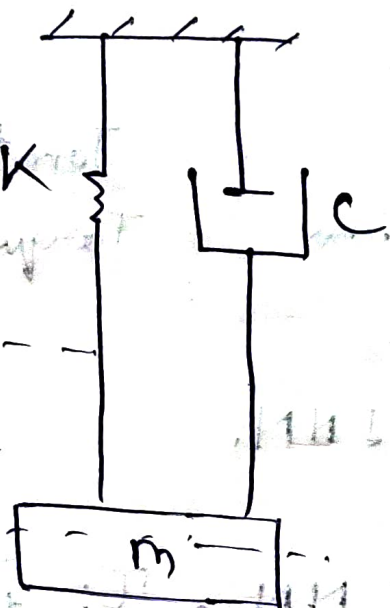
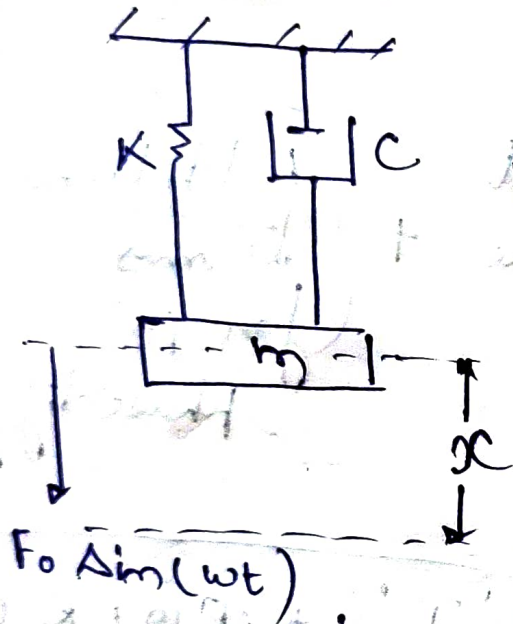
→ Friction.

Harmonic drive model



Spring-Damper System

(For linear translation) (33)



$$\begin{aligned} x &= r \theta \\ \dot{x} &= r \dot{\theta} \\ \ddot{x} &= r \ddot{\theta} \end{aligned}$$

$$\begin{aligned} \theta &= \frac{x}{r} \\ \dot{\theta} &= \frac{\dot{x}}{r} \\ \ddot{\theta} &= \frac{\ddot{x}}{r} \end{aligned}$$

$$m \ddot{x} + C \dot{x} + K x = F_0 \sin(wt)$$

$$m \ddot{\theta} + C r \dot{\theta} + K r \theta = F_0 \sin(wt)$$

→ Multiply by 'r' both sides

$$\tau = F_0 r$$

$$\rightarrow m r \ddot{\theta} + C r^2 \dot{\theta} + K r^2 \theta = F_0 r \sin(wt)$$

$$\rightarrow J \ddot{\theta} + C_t \dot{\theta} + K_t \theta = \tau \sin(wt)$$

For rotational motion (revolute joint). (34)

Motor Torque + corrections = Joint Torque + Additional terms (for 90° phase difference)
 $\downarrow$
 by PINN.

$$\Rightarrow \underbrace{N \tau_m + \dots}_{\text{Applied external force}} = \tau_j + \underbrace{J \ddot{\theta}}_{\text{Static friction}} + \underbrace{B \dot{\theta}}_{\text{Viscous}} + \underbrace{\tau_c \tanh\left(\frac{v}{v_s}\right)}_{\text{Coulomb}}$$

From NLM and law:-

$$F = \frac{d(P)}{dt}$$

$$\tau_{\text{net}} = I \alpha = J \ddot{\theta} = J \ddot{\theta}(t)$$

(overall) Rotor inertia.
~~overall rotor~~

And: virtual torques on rotor;

Driving torque = $N \cdot \tau_m$
~~Viscous fric~~
 Viscous fric = $-B \cdot \dot{\theta} = -B \cdot \dot{\theta}(t)$
 Coulomb " = $-\tau_c \tanh\left(\frac{v}{v_s}\right)$; $v = \dot{\theta}(t)$
 Reaction torque = $-\tau_j$

$$J_m \ddot{\theta} = N T_m - B \dot{q}(t) - T_c \tanh\left(\frac{v}{v_s}\right) - \tau_j$$

→ ~~QERO~~

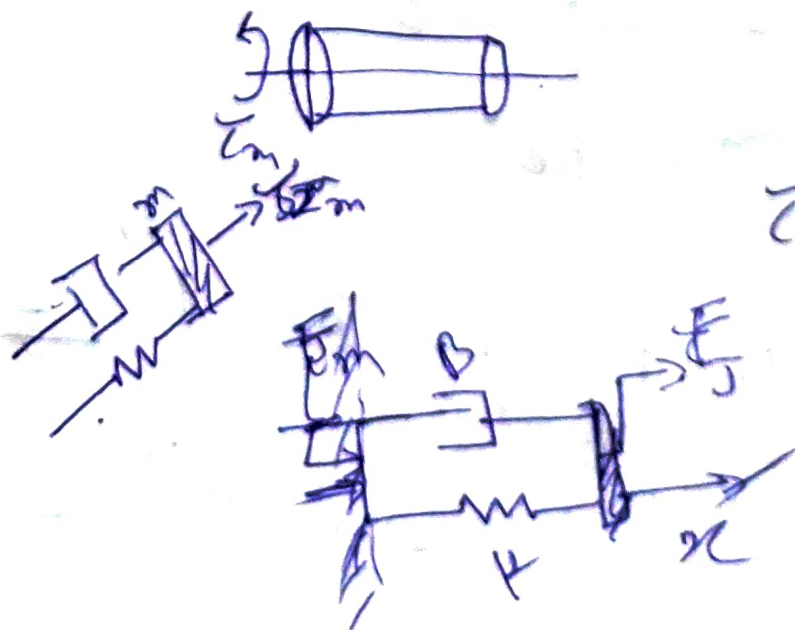
$$T_j = N T_m - J_m \ddot{\theta} - B \dot{q}(t) - T_c \tanh\left(\frac{v}{v_s}\right) + T_o + f_{NN}(\text{INN})$$

↓
Static friction.

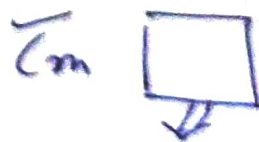
Lagrangian eq = KE - PE.

$$L(\theta, \dot{\theta}) = K(\theta, \dot{\theta}) - P(\theta).$$

$$\tau = \frac{d}{dt} \cdot \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta}.$$

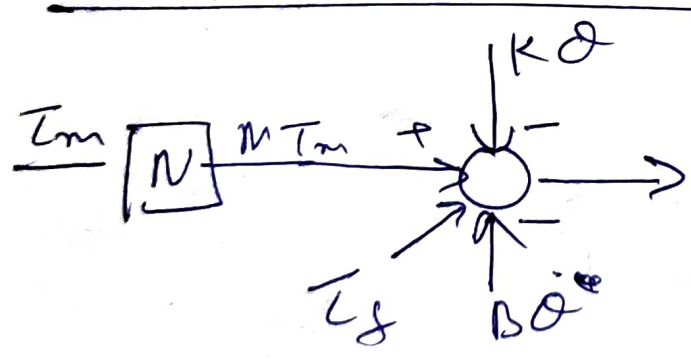
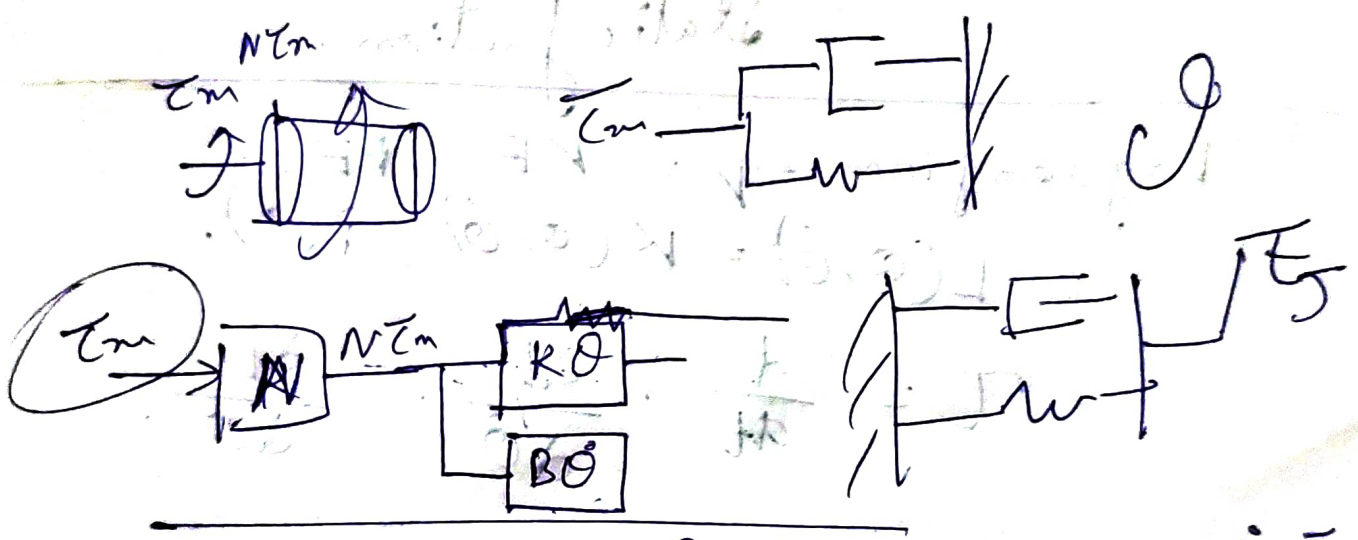
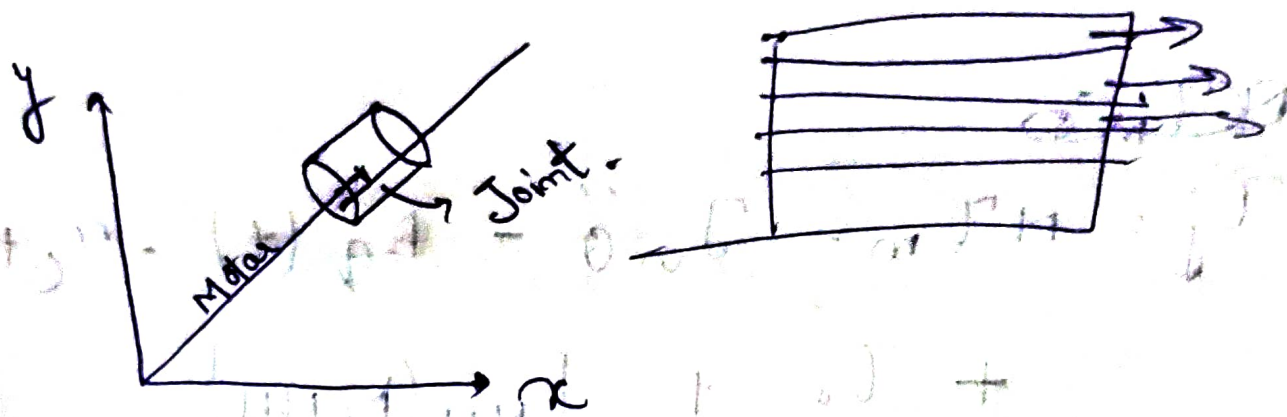


$$\tau_m = \tau_j$$



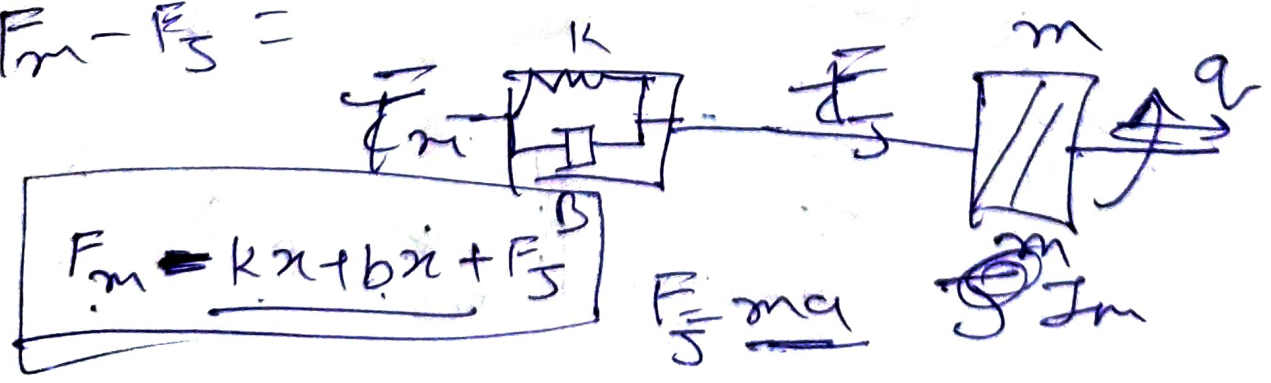
$$m \ddot{x} + B \dot{x} + kx = F_j$$

$$m \ddot{x} + B \dot{x} + kx = F_j - F_m$$



$$N\tau_m - K\theta - B\dot{\theta} - \tau_s = \tau_s$$

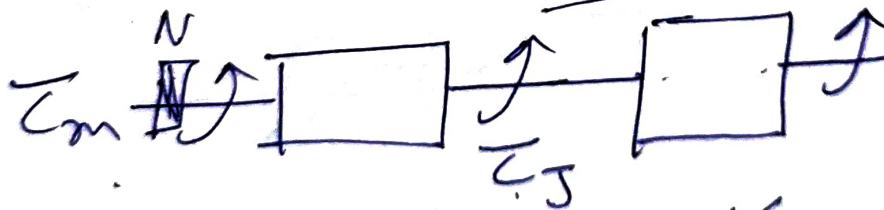
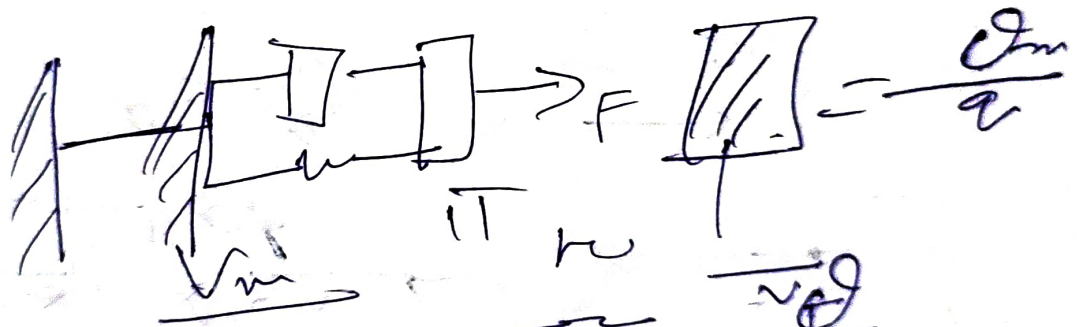
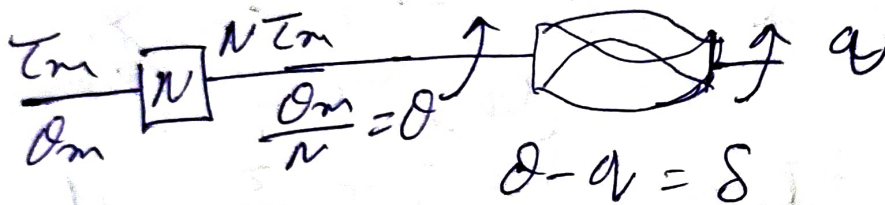
$$F_m - F_s =$$



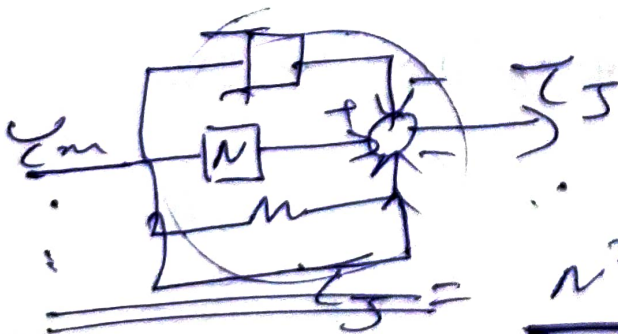
(37)

$$\delta = \theta - q = \frac{\theta_m}{N} - q$$

$$\tau_J = D_n \dot{\delta} + K_n \delta$$



$$\tau_J = I_L \ddot{\theta} + B \dot{\theta} + K \theta$$



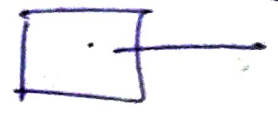
$$N \tau_m - B \dot{\theta} - K \theta - I_L \ddot{\theta} = \tau_J$$

$$\tau_J = \frac{N^2 I_L}{\tau_J} \ddot{\theta} + B \dot{\theta} + K \theta + K \tau_J$$

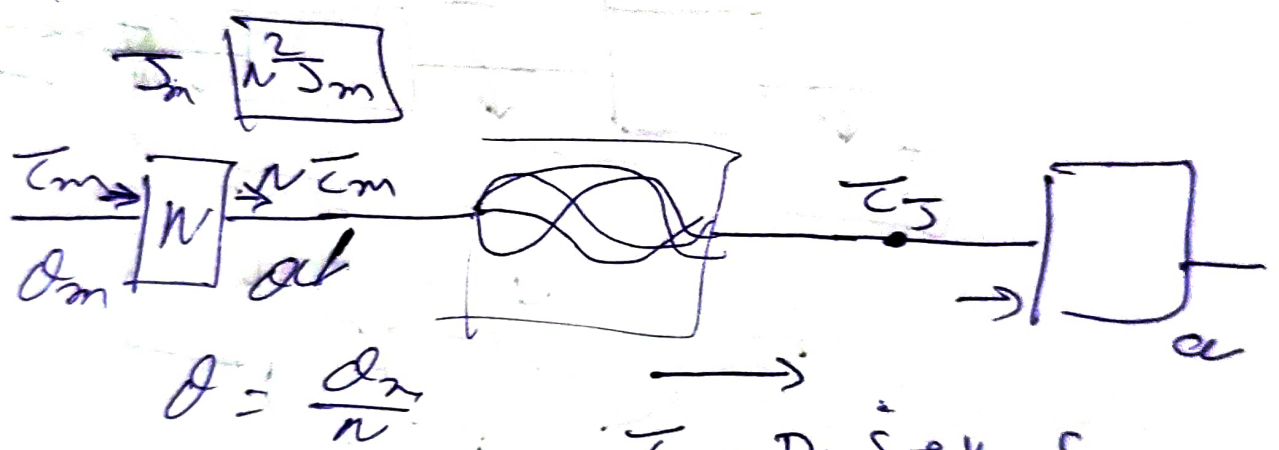
(1.2)

$$\tau_j = k_n \delta + D_n \dot{\delta}$$

$$\delta = \frac{\theta_m}{n} - a$$



$$J_L \ddot{a} + B_L \dot{a} + \tau_{load} = \tau_j$$



$$K_m = \frac{1}{2} J_m \dot{\theta}_m^2$$

$$= \frac{1}{2} J_m (n \dot{\theta})^2$$

$$\underline{K_m = \frac{1}{2} J_m n^2 \dot{\theta}^2}$$

$$\tau_j = D_n \dot{\delta} + k_n \delta$$

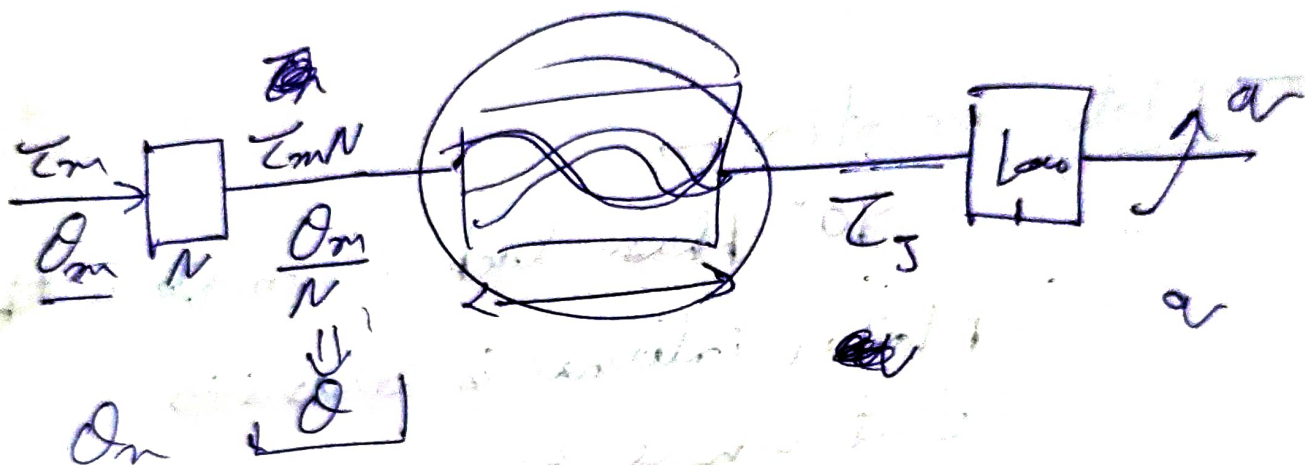
$$\tau_j = J_L \ddot{a} + \dots$$

$$D_n \dot{\delta} + k_n \delta = J_L \ddot{a} + B_L \dot{a} + \tau_{load}$$

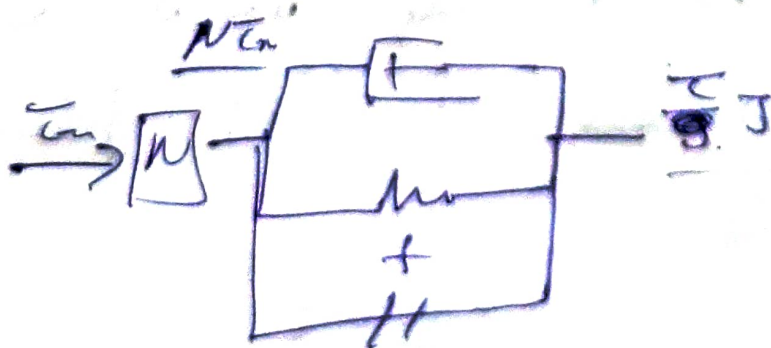
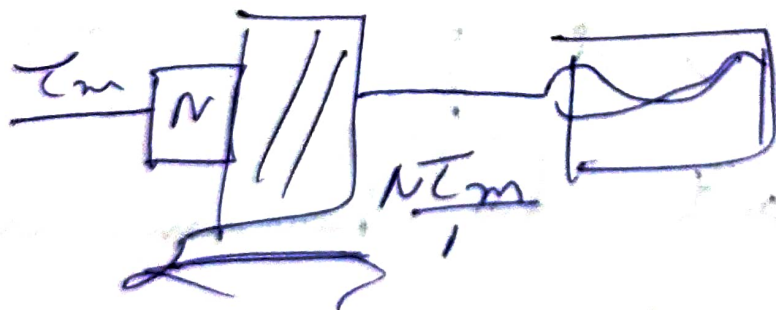
$$D_n \left(\frac{\dot{\theta}_m}{n} - \dot{a} \right) + k_n \left(\frac{\theta_m}{n} - a \right) = J_L \ddot{a} + B_L \dot{a} + \tau_{load}$$

29

Diagram showing a signal path with a feedback loop and a summing junction.



U_m



→ Motor inertia →

90° phase lock in motor with drive
Motor internal is elastic,
but whole sys is inelastic.

→ $T_m = N^2 J_m \ddot{\theta} + \dots$ Add flex model.
 ↳ $\theta = q$, $\dot{\theta} = \dot{q}$

so we didn't do harmonic drive
modelling but instead we are able
to find: ~~assume~~ find 90° phase
shift in motor.
we are not taking dynamics b/w a & q.

$$\frac{x_0}{x} = \frac{10}{20}$$

$$\frac{T_c}{V_s}$$

$$\frac{V_c}{V_s}$$

$$T_c = \frac{20}{V_s} = f\left(\frac{x_0}{x}\right)$$

$$V_s = 10$$

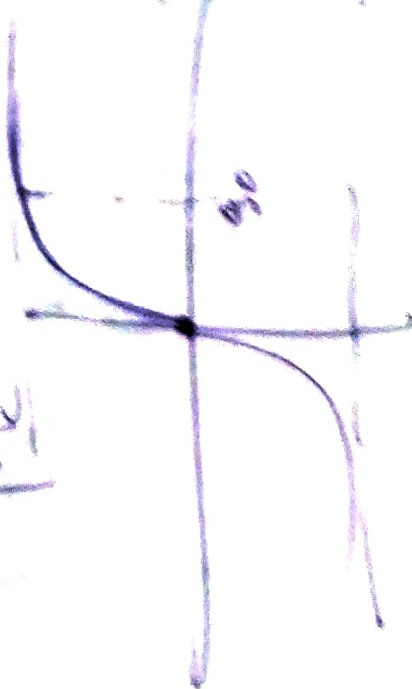
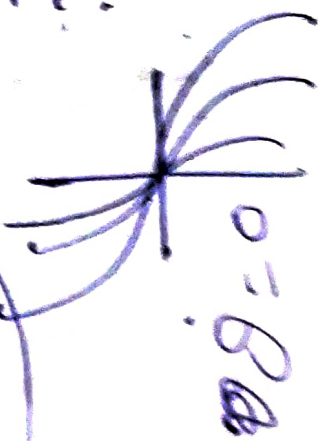
$$T_c = 10$$

$$V_s = 20$$

$$T_c = 5$$

$$T_c = 10$$

$$T_c \times (-1, 1)$$



$$V_s = 10$$



$$V_s = 20$$

$$\frac{T_c}{V_s} = \text{const}$$

$$T_m = T_i + N^2 J_m \ddot{\theta} + N^2 B_m \dot{\theta}$$

$$J_m < \dots$$

$$B_m < \dots$$

$$T_c = \dots$$



$$-1 < \dots$$

(42)

$$\underline{N\tau_m + N\tau_j = \tau_j \pm \Phi (90^\circ)}$$

~~Force~~

→ Power

$$\tau_m \rightarrow 0$$

$$\underline{\tau_f = N\tau_m - \tau_j \pm 90^\circ}$$



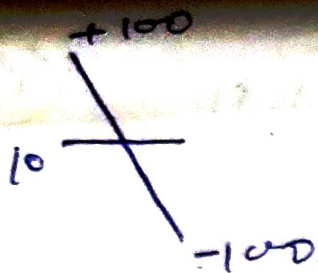
$$\tau_c \tanh\left(\frac{v}{v_s}\right)$$

$$\tau_c, v_s$$

JST → 2 freq.

High loss $\rightarrow$ 2 ter.

we don't have α_{meter} , → So $k = t(\alpha_m - \alpha_j)$
 So I need to do only Block here $N\tau$.



$$N I_m - I_g = I_g$$

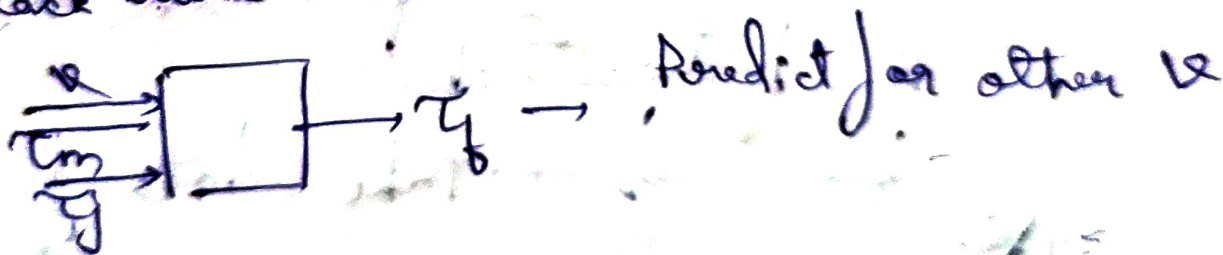
$$N I_m - I_g = I_c \tanh\left(\frac{V}{V_S}\right)$$

$\downarrow$ $\downarrow$ Phy
 100 = (I_c, V_S)

Tasks.

12/06/26

① Do black box NN model



② Physics based model.

$$100 I_m - I_g = I_c \cdot \tanh\left(a + \frac{V}{V_S}\right)$$

Predict $\swarrow$ I_c, a, V_S .

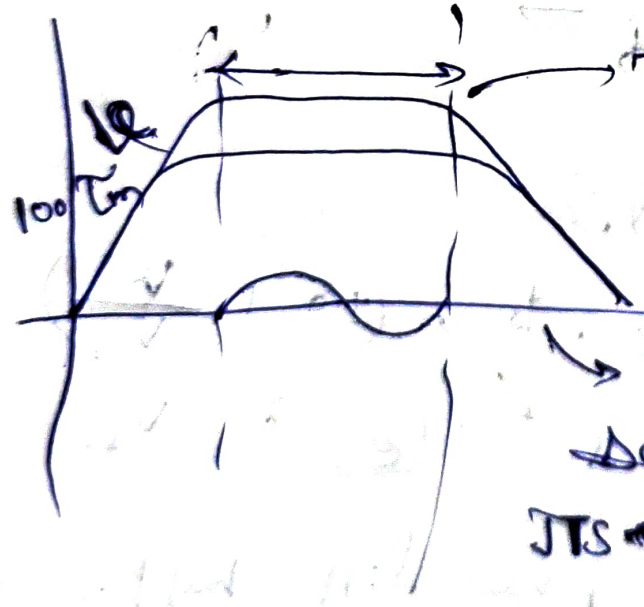
③ Study why the phase diff happens in JTS & AITS.
Like control sys dynamics.

(44)

Like why there is a phase diff
b/w JST ~~MT~~ and da
Root Mean Square ^{Value} of JTS, will it work
100 x MT $\rightarrow$ Phase same as velocity
why.

(4)

Da Laplace transformation of the
eq, pictures taken.



Take data for this
seg only for
Laplace transformation

$\rightarrow$ Have can lie in
same phase and
JTS ~~MT~~ in other phase.

Observations

- 1) ~~Motor~~ T_m and vel ~~for~~ nearly same phase.
 Servo controll motor

Interpretation $T_m = J_{eq} \ddot{\theta} + B \dot{\theta} + T_f + T_{load}$

→ Vel constant since trapezoidal trajectory.

$$\ddot{\theta} \approx 0 \quad \text{and} \quad T_m \propto \dot{\theta}$$

→ During vel controll experiment, servo control generates motor torque primarily to achieve commanded velocity.

∴ T_m and vel appear correlated.

→ Motor current loop is fast (kHz range), torque generation follows vel command immediately.

How to estimate friction parameters, $\rightarrow$ Iterative method. (46)

$\rightarrow$ Non linear least square estimation implemented through SciPy's curve_fit() package.

$$\rightarrow \check{T}_f = T_c \tanh\left(\frac{\check{v}}{v_s}\right) + B_v \times \check{v}$$

$\rightarrow$ Unknown $\Rightarrow$ T_c
 v_s
 B_v

$\rightarrow$ Initially, $T_c = 5$, $B_v = 0.1$, $v_s = 0.01$

$$\hat{T}_f \text{ predicted} = \hat{T}_f = T_c \tanh\left(\frac{v}{v_s}\right) + B_v \times v$$

$\rightarrow$ Predict not perfect.

$$\rightarrow \text{Error} = T_f - \hat{T}_f$$

$$\rightarrow \text{Minimise} \rightarrow (\text{error})^2 = \sum_{i=1}^N (T_f - \hat{T}_f)^2$$

$\rightarrow$ Iterative method $\rightarrow$ 1st $\rightarrow T_c, B_v, v_s$

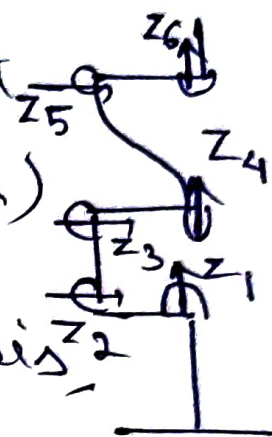
$\rightarrow$ 2nd, $T_c = 5.3$, $B_v = 0.13$, $v_s = 0.015$

$\rightarrow$ 3rd, $T_c = 5.8$, $B_v = 0.15$, $v_s = 0.020$

$\rightarrow$ Error² min.

Report (Presentation in 1 hour)

- Literal Introduction → What is what
- Literature overview, (Justification)
- Robotic arm design and analysis



- Find kinematics, DH parameters
- Get DH parameter → ACW thumb (+ve)
- What exp covered out (details)
- What is the main goal.
- why this exp, justification and acc to existing literature.
- What ML model used step by step, why this
- At initial stages only — coloum + viscous done later we can go to non linear eq, static check.
- I have only done kinematics, not the dynamics since by seeing from outside cannot get idea about internal structure.